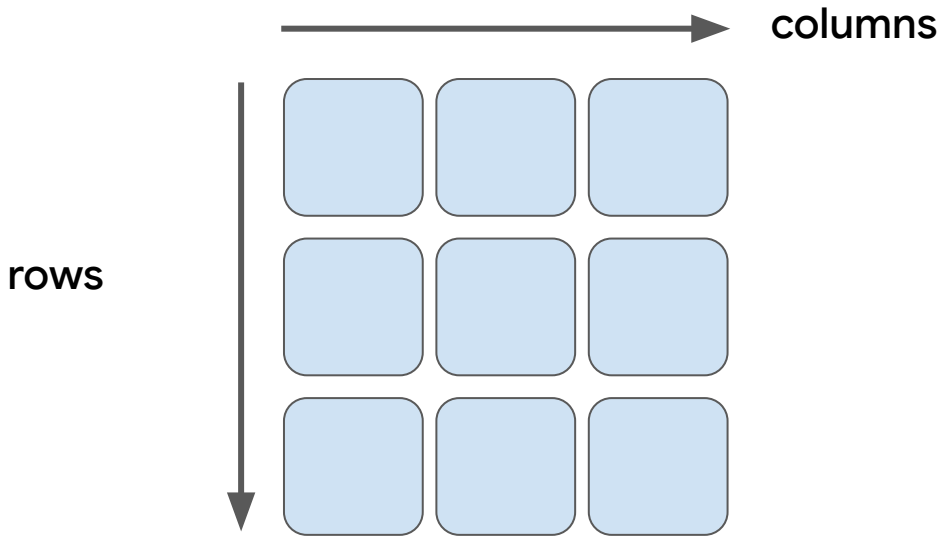
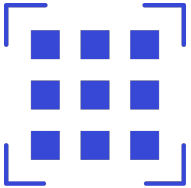
Matrices

Matrices

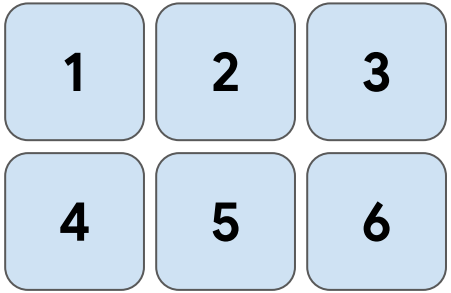
- Two-dimensional arrays can be defined as arrays within an array.
- 2D arrays erected as matrices, which is a collection of rows and columns.

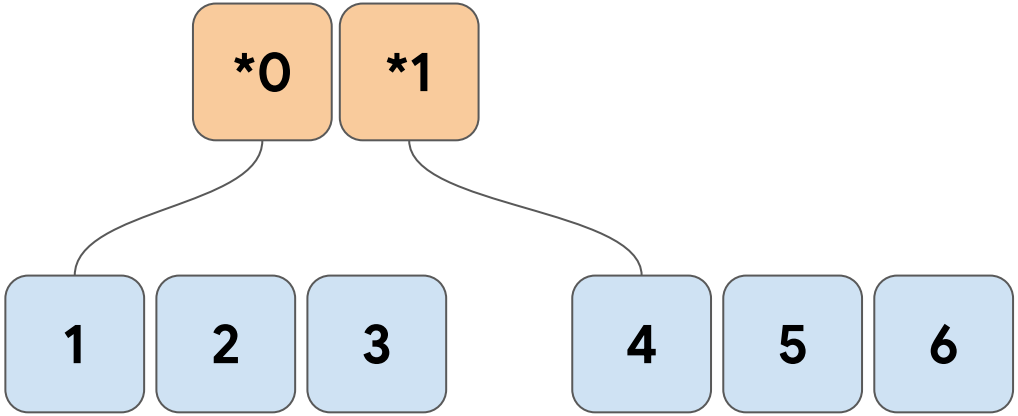


Matrices

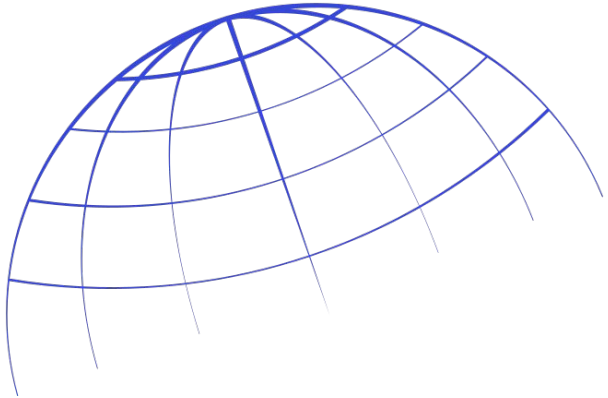


- You should think two dimensional arrays as matrices.
- In reality, they are arrays holding references to other arrays

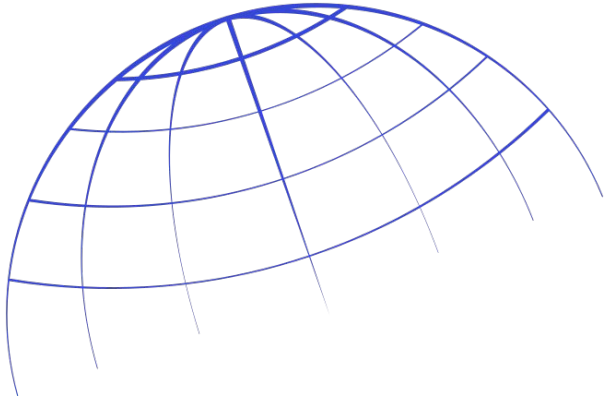




Common Approaches



Traversing Matrices



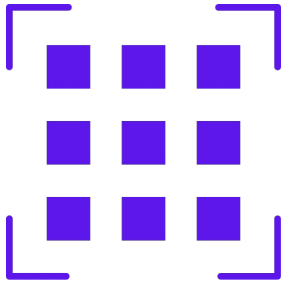
Traversing - from Top-left to Bottom-right

Approach

- Start from (0, 0)
- Increment column index by one for each row you visit

Implementation

```
for row_idx in range(len(matrix)):
    for col_idx in range(len(matrix[0])):
        print(matrix[row_idx][col_idx])
```



Traversing - From Bottom-right to Top-left

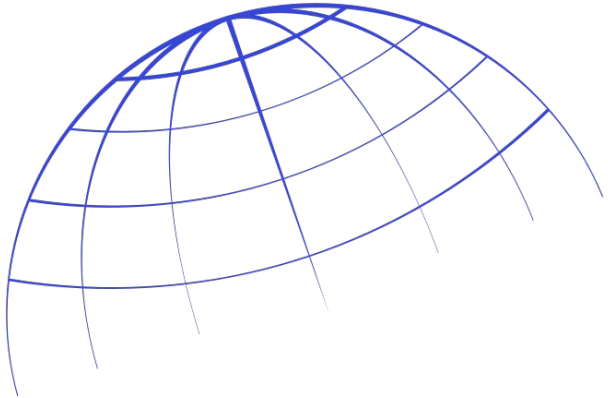
Approach

- Start from (`last_row - 1`, `last_col - 1`)
- Decrement column index by one for each row you visit.

Implementation

```
for row_idx in range(len(matrix) - 1, -1, -1):
    for col_idx in range(len(matrix[0]) - 1, -1, -1):
        print(matrix[row_idx][col_idx])
```

Enumerating Cells/ 2D - 1D Mapping



Enumerating Cells - Approach

	0	1	2	3
0	0	1	2	3
1	4	5	6	7
2	8	9	10	11

Enumerating Cells - Approach

Consecutive elements in a column differ by the number of columns

	0	1	2	3
0	0	1	2	3
1	4	5	6	7
2	8	9	10	11

Enumerating Cells - Approach

Consecutive elements in a row differ by 1

	0	1	2	3
0	0	1	2	3
1	4	5	6	7
2	8	9	10	11

Enumerating Cells - Approach

Therefore the 1D number of a cell is `row_number*n_columns + column_number`

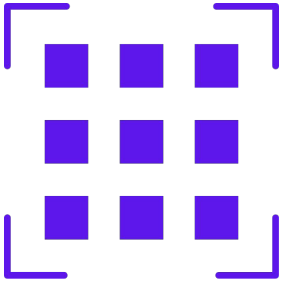
	0	1	2	3
0	0	1	2	3
1	4	5	6	7
2	8	9	10	11

Enumerating Cells - Remainder Theorem

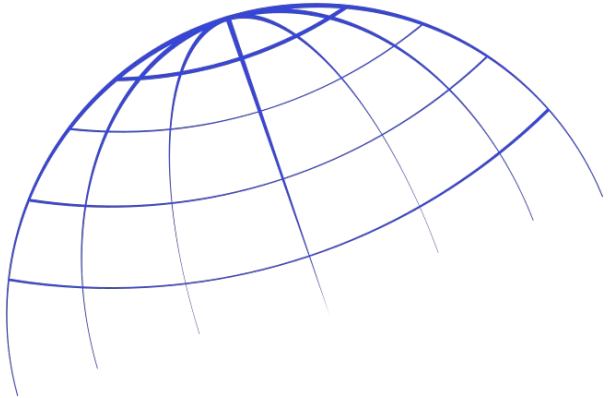
```
cell_number = row_number*n_columns + column_number

row_number = cell_number // n_columns

column_number = cell_number % n_columns
```



Diagonal Keys



Diagonal Keys

What is common about the highlighted column and row numbers?

	0	1	2	3
0	0	1	2	3
1	4	5	6	7
2	8	9	10	11

Diagonal Keys

What is common about the highlighted column and row numbers?

	0	1	2	3
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1	4	5	6	7
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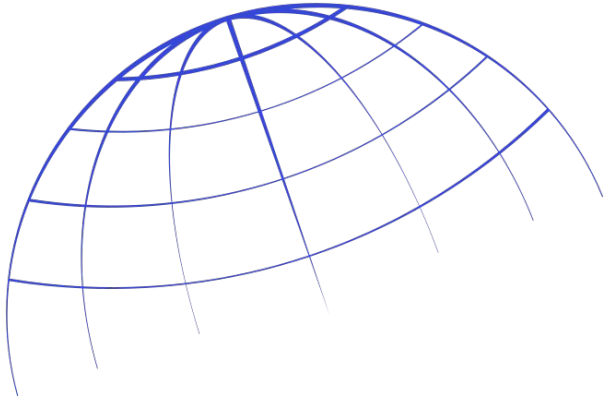
Common Pitfalls



Common Pitfalls - Index Out of Bound

- Trying to access elements using indices that are greater (or equal to) than the size of our array will result in this exception.

Testcase **Result**

```
IndexError: list index out of range
    print(nums[i])
Line 4 in twoSum (Solution.py)
    ret = Solution().twoSum(param_1, param_2)
Line 28 in _driver (Solution.py)
    _driver()
Line 39 in <module> (Solution.py)
```

Stdout

```
2
```

Console Run Submit

Common Pitfalls - Negative Indexing

- Because we may use negative indices in Python, it is a typical problem to use negative indices accidently, which can result in a highly troublesome misunderstanding if no exception is triggered.

Negative indices:	-4	-3	-2	-1
Positive indices:	0	1	2	3
Values:	`[1,`	`2,`	`3,`	`4]`

Common Pitfalls - Copying lists

- A list is a reference. Take care when you want to you want to pass only the content of the list.

```
nums[::]
nums.copy()
```

Common Pitfalls - Initializing with *

- The * operator can be used as `[object]*n` where n is the number of elements in the array.
- In the case of 2D arrays, this will result in shallow lists
- Hence using list comprehensions is a safer way to create 2D lists.

```
row = len(matrix)
col = len(matrix[0])

transposed  = [[0]*rows for _ in range(cols)]
```

Common Pitfalls - Using `insert`, `pop` and `in`

- insert - inserts a given element at a given index
  - `insert(i, value)` : O(N)
- pop - returns the value at a given index and removes it from the list where the default is the last index
  - `pop` : O(1)
  - `pop(i)` : O(n)

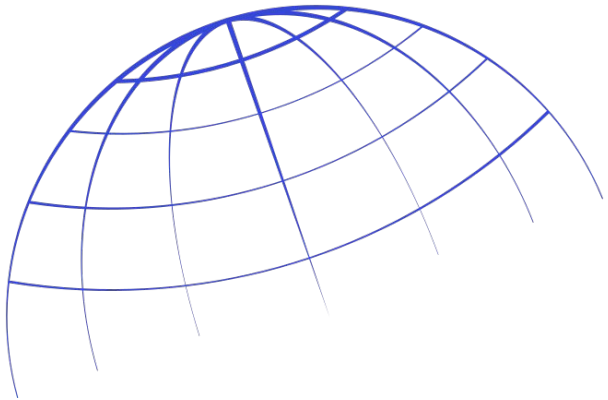
- The `in` operator returns a boolean denoting whether an element is present in the iterable data structure we use it on.
  - Lists O(N)
  - Map, Set O(1)

Common Pitfalls - Mistaking Row Iterators with Column iterators

- It is pretty common to mistake your `i` and `j`'s representing rows and columns. And sometimes, especially when the matrix is a square one, it does not throw an exception but hours of confusion

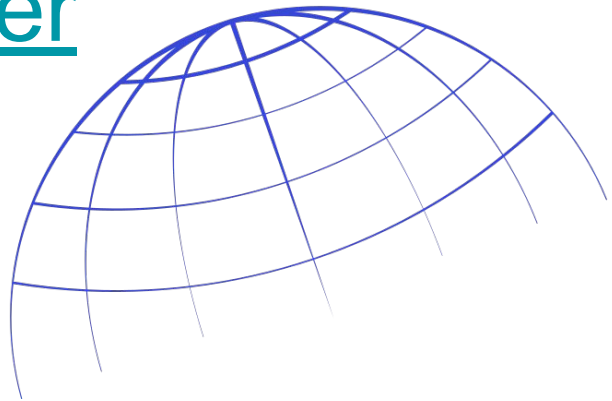
```
for col_idx in range(len(matrix[0])):
    for row_idx in range(len(matrix)):
        print(matrix[col_idx][row_idx])
```

Checkpoint



Pair Programming Question

Image smoother



Practice Questions

Reverse string

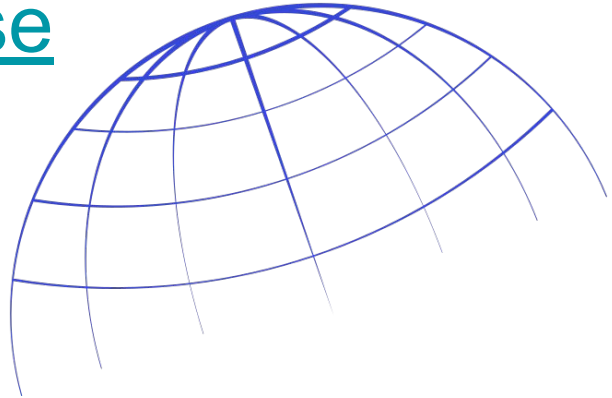
Count Equal and Divisible Pairs in an..

All Divisions With the Highest Score..

Transpose matrix

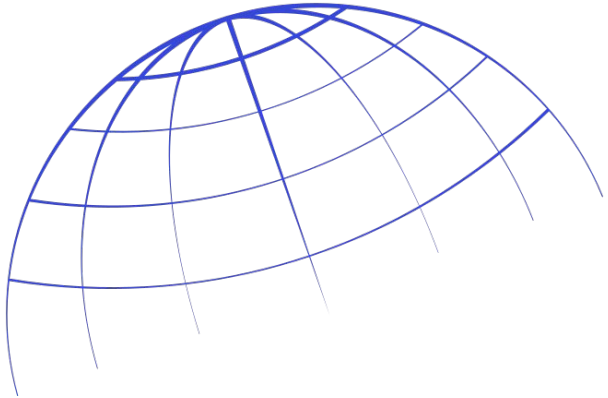
Diagonal traverse

Rotate image

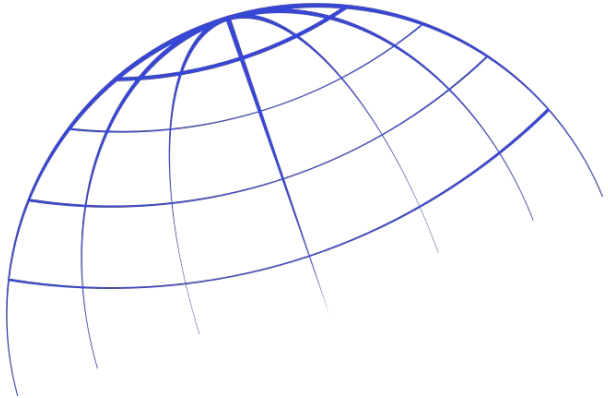


Resources

- https://neetcode.io/courses/dsa-for-beginners/0: useful on laying out the fundamentals
- https://www.geeksforgeeks.org/python-arrays/: covers large range of topics for python
- Python List pop() Method - GeeksforGeeks: good examples
- Python List insert() Method With Examples - GeeksforGeeks: good examples
- Python | Which is faster to initialize lists? - GeeksforGeeks: good article on list initialization



Quote of the Day

If you want to enjoy the rainbow,
be prepared to endure the storm

Warren W. Wiersbe